Wan technologies assignment

Name -: harsh oza

1. Wan vs lan comparison table .

Feature	LAN (Local Area Network)	WAN (Wide Area Network)
Full Form	Local Area Network	Wide Area Network
Coverage Area	Small area (home, office, building)	Large area (city, country, worldwide)
Ownership	Usually owned by one organization	Often uses public/private telecom networks
Speed	High speed (100 Mbps to several Gbps)	Generally slower than LAN
Cost	Low setup and maintenance cost	Higher setup and maintenance cost
Data Transfer Rate	Fast	Comparatively slower
Latency	Low latency	Higher latency
Security	More secure and easier to manage	Less secure, requires additional security measures
Examples	Office network, school network, home Wi-Fi	Internet, company networks connecting multiple branches
Devices Used	Switches, Access Points, Routers	Routers, Modems, Leased Lines, MPLS links
Reliability	High reliability	Depends on service provider and network links
Management	Easy to manage	More complex to manage

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2. Network diagram.

```
Ahmedabad User
            |
        [Laptop/PC]
            |
        [Switch]
            |
        [Router]
            |
    ======= WAN =======
    (Internet / ISP Network)
            |
            |
        [Router]
            |
      [Core Switch]
            |
    Flipkart Data Center
        Bangalore
            |
      [Web Servers]
            |
      [Database Servers]

  <------ LAN ------>     <------ LAN ------>
   Ahmedabad Office        Flipkart Data Center
```

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3 . Research and write a short summary (5-7 lines) explaining what MPLS is and how it improves data routing compared to traditional IP routing.

MPLS (Multiprotocol Label Switching) is a high-performance networking technology that directs data using short labels instead of performing complex IP address lookups at every router. When a packet enters an MPLS network, a label is attached to it. Routers inside the MPLS network forward packets based on these labels, making routing faster and more efficient. MPLS can also prioritize important traffic such as video streaming, VoIP, and online transactions. Compared to traditional IP routing, MPLS reduces delays, improves traffic management, and provides better Quality of Service (QoS).

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4. **Observe the pre-configured MPLS network in the provided Packet Tracer or GNS3 file, and note down two routing behaviors that are different from standard IP routing.

Hint: Look for how labels are used and how packets move between routers.**

Label-Based Forwarding

- In MPLS, routers forward packets using labels rather than checking the destination IP address at every hop.
- In traditional IP routing, each router examines the IP header and performs a routing table lookup.

Label Swapping

- MPLS routers replace (swap) the incoming label with a new label before forwarding the packet.
- In standard IP routing, packets keep the same IP header while routers simply forward them based on routing table entries.

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5. Three Benefits of MPLS for High-Traffic Applications

Benefit	Explanation	Real-World Example
Faster Packet Forwarding	MPLS uses labels instead of repeated IP lookups, reducing processing time.	YouTube can deliver videos with lower latency and smoother playback.
Traffic Engineering	MPLS allows network administrators to choose specific paths for different traffic types.	BookMyShow can route ticket-booking traffic through less congested paths during major movie releases.
Quality of Service (QoS)	MPLS can prioritize critical applications over less important traffic.	Video streaming, online payments, and live event bookings receive higher priority, improving user experience.

End

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